Transport in plants and humans

Subject: Biology | Level: Class 10 | Curriculum: Federal Board
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1. Topic Overview and Learning Objectives

The study of transport in living organisms is fundamental to understanding how life is sustained through the constant movement of nutrients, gases, and waste products. In plants, this involves a sophisticated system of vascular tissues, namely xylem and phloem, which operate based on physical principles like cohesion, tension, and osmotic pressure. For Class 10 students, mastering these concepts is essential as they form the basis for more advanced botanical studies and frequently appear in the long-answer sections of the Federal Board biology paper.

In humans, the transport system is centered around the circulatory system, which includes the heart, blood vessels, and blood. This system is a marvel of biological engineering, ensuring that every cell in the body receives oxygen and nutrients while efficiently removing metabolic wastes like carbon dioxide. The Federal Board curriculum emphasizes the structure of the heart, the mechanics of the cardiac cycle, and the composition of blood. Understanding these topics is not only vital for scoring high marks but also for developing a foundational knowledge of human health and common medical conditions.

The intuition behind transport systems lies in the concept of efficiency and distribution. Just as a city requires a network of roads and pipes to function, complex multicellular organisms need specialized pathways to bridge the gap between resource intake and cellular utilization. By studying the similarities and differences between plant and animal transport, students can appreciate the diverse evolutionary strategies used to solve the problem of internal distribution. These notes provide a structured approach to revising these critical topics, ensuring you are well-prepared for both objective and subjective exam questions.

2. Core Concepts and Detailed Revision Notes

2.1 Mechanism of Water and Food Transport in Plants

Transport in plants is a fascinating process that defies gravity through specialized vascular tissues. It primarily involves the movement of water and minerals from the roots to the leaves through the xylem, and the distribution of synthesized food from the leaves to other parts of the plant via the phloem. The most widely accepted mechanism for water transport is the Cohesion-Tension Theory. This theory suggests that the evaporation of water from leaves, known as transpiration, creates a negative pressure or tension that pulls water upward. Because water molecules are cohesive, meaning they stick to each other, and adhesive, meaning they stick to xylem walls, they form a continuous column that can reach the tallest trees. Meanwhile, the Pressure-Flow Hypothesis explains food transport, where sugar is actively loaded into phloem sieve tubes, creating an osmotic gradient that moves sap from source to sink. Understanding these mechanisms is crucial for mastering plant physiology and answering board exam questions on plant survival.

Key Points to Remember:

1. Xylem transports water and minerals upward from roots to leaves using transpiration pull.
2. Phloem transports organic nutrients (sucrose) from leaves to all parts of the plant.
3. Transpiration is the loss of water vapor from the aerial parts of the plant, mainly through stomata.
4. The Cohesion-Tension Theory explains how water moves in a continuous column without breaking.
5. The Pressure-Flow Hypothesis describes the movement of food from source (leaves) to sink (roots/fruits).

2.2 Structure and Function of the Human Heart

The human heart is a remarkable muscular organ that acts as a double pump, ensuring that oxygenated and deoxygenated blood remain separate. It consists of four chambers: two upper atria and two lower ventricles. The right side of the heart receives deoxygenated blood from the body via the vena cava and pumps it to the lungs through the pulmonary artery. Conversely, the left side receives oxygen-rich blood from the lungs via the pulmonary veins and pumps it to the rest of the body through the aorta. This separation is maintained by a thick muscular wall called the septum. Valves, such as the tricuspid and bicuspid (mitral) valves, play a critical role in preventing the backflow of blood, ensuring a one-way circulation. The rhythmic contraction known as systole and relaxation known as diastole constitute the cardiac cycle, which is essential for maintaining life and consistent blood pressure. Students must focus on the pathway of blood flow and the specific functions of each valve for the board exam.

Key Points to Remember:

1. The heart has four chambers: right atrium, right ventricle, left atrium, and left ventricle.
2. The right side handles deoxygenated blood, while the left side handles oxygenated blood.
3. Valves like the tricuspid, bicuspid, and semi-lunar valves ensure one-way blood flow.
4. The cardiac cycle consists of atrial systole, ventricular systole, and complete cardiac diastole.
5. The pulmonary circuit goes to the lungs, while the systemic circuit goes to the rest of the body.

2.3 Composition and Functions of Human Blood

Blood is a specialized fluid connective tissue that serves as the primary medium of transport in the human body. It consists of a liquid component called plasma and various formed elements, including red blood cells (erythrocytes), white blood cells (leukocytes), and platelets (thrombocytes). Plasma makes up about 55 percent of the blood volume and carries dissolved nutrients, hormones, proteins, and waste products. Red blood cells are biconcave discs packed with hemoglobin, a protein that binds to oxygen for delivery to tissues. White blood cells are the soldiers of the immune system, defending the body against pathogens through phagocytosis or antibody production. Platelets are essential for

blood clotting, preventing excessive bleeding after an injury by forming a plug at the site of vascular damage. Each component has a specific, vital role, and understanding their proportions and functions is a common topic in the Federal Board biology syllabus, often appearing in both multiple-choice and short-answer questions.

Key Points to Remember:

1. Plasma is the liquid part of blood containing water, salts, and proteins.
2. Red Blood Cells (RBCs) contain hemoglobin for oxygen transport and have no nucleus when mature.
3. White Blood Cells (WBCs) are involved in the body's immune response and defense mechanisms.
4. Platelets are cell fragments that initiate the blood clotting process to prevent blood loss.
5. Blood also helps in temperature regulation and maintaining the pH balance of the body.

2.4 Cardiovascular Disorders and Prevention

Cardiovascular disorders are a significant health concern and a frequent topic in biology examinations. These conditions affect the heart and blood vessels, often leading to serious complications if left untreated. Atherosclerosis is a common condition where fatty deposits called plaques build up inside the arteries, narrowing the lumen and restricting blood flow. If these plaques harden due to calcium deposition, the condition is known as arteriosclerosis, which makes the vessel walls rigid. These issues can lead to hypertension, which is chronically high blood pressure, and significantly increase the risk of a myocardial infarction, commonly known as a heart attack. A heart attack occurs when the blood supply to a part of the heart muscle is blocked, leading to tissue death. Understanding the risk factors, such as high-fat diets, smoking, obesity, and lack of exercise, is essential for both exam preparation and personal health awareness. Students should be able to distinguish between these terms clearly to avoid confusion in the exam.

Key Points to Remember:

1. Atherosclerosis involves the buildup of fatty plaques in the internal layers of arteries.
2. Arteriosclerosis is the hardening of the arteries, often following atherosclerosis.
3. Hypertension is a condition of persistent high blood pressure (above 140/90 mmHg).
4. Myocardial infarction occurs when coronary arteries are blocked, damaging heart muscle.
5. Prevention includes a balanced diet, regular exercise, and avoiding tobacco products.

3. Key Definitions, Formulas, and Rules

1. **Transpiration:** The loss of water from the plant surface in the form of water vapor, primarily through stomata.
2. **Xylem:** Vascular tissue responsible for the conduction of water and minerals from roots to leaves.
3. **Phloem:** Vascular tissue responsible for the translocation of organic food from leaves to other parts.
4. **Systole:** The phase of the cardiac cycle when the heart muscle contracts and pumps blood into the arteries.
5. **Diastole:** The phase of the cardiac cycle when the heart muscle relaxes and the chambers fill with blood.
6. **Hemoglobin:** An iron-containing protein in red blood cells that binds and transports oxygen.
7. **Antigen:** A specific protein or carbohydrate on the surface of red blood cells that determines blood type.
8. **Agglutination:** The clumping of red blood cells that occurs when incompatible blood types are mixed.
9. **Pulse:** The rhythmic expansion and contraction of an artery as blood is pumped through it by the heart.
10. **Double Circulation:** A circulatory system where blood passes through the heart twice in one complete circuit (Pulmonary and Systemic).

4. Step-by-Step Worked Examples

Example 1: Calculating Transpiration Rate

Problem: A plant loses 50 ml of water over a period of 5 hours. Calculate the rate of transpiration in ml per hour.

Solution Steps:

Step 1: 1. Identify the total volume of water lost (50 ml).

Step 2: 2. Identify the total time taken (5 hours).

Step 3: 3. Use the formula: Rate = Total Volume / Total Time.

Step 4: 4. Divide 50 by 5.

Final Answer: The rate of transpiration is 10 ml/hour.

Example 2: Determining Blood Group Compatibility

Problem: A patient with Blood Group A negative needs a transfusion. Which blood groups are compatible for this patient?

Solution Steps:

Step 1: 1. Identify the antigens present: Only Antigen A is present on the RBCs.

Step 2: 2. Identify the antibodies in plasma: Anti-B and Anti-Rh (if sensitized).

Step 3: 3. Check for Rh factor: The patient is Rh negative, so they cannot receive Rh positive blood.

Step 4: 4. List compatible donors: Group A negative and Group O negative (universal donor).

Final Answer: The compatible blood groups are A negative and O negative.

Example 3: Tracing Blood Flow through the Heart

Problem: Describe the path of a red blood cell from the superior vena cava to the lungs.

Solution Steps:

Step 1: 1. The RBC enters the Right Atrium from the Superior Vena Cava.

Step 2: 2. It passes through the Tricuspid Valve into the Right Ventricle.

Step 3: 3. The Right Ventricle contracts, pushing the RBC through the Pulmonary Semi-lunar Valve.

Step 4: 4. The RBC enters the Pulmonary Artery which carries it to the lungs.

Final Answer: Vena Cava -> Right Atrium -> Tricuspid Valve -> Right Ventricle -> Pulmonary Artery -> Lungs.

Example 4: Identifying Vessel Types

Problem: A blood vessel has thick elastic walls and carries blood under high pressure away from the heart. Identify the vessel type.

Solution Steps:

Step 1: 1. Observe the direction of flow: Away from the heart indicates an artery.

Step 2: 2. Observe the wall structure: Thick and elastic walls are characteristic of arteries to withstand pressure.

Step 3: 3. Observe the pressure level: High pressure is found in the systemic arterial system.

Final Answer: The blood vessel is an Artery.

5. Common Exam Mistakes to Avoid

1. Confusing the direction of transport in xylem and phloem; remember Xylem is Up (water) and Phloem is Flow (food in all directions).
2. Stating that all arteries carry oxygenated blood; always mention the exception of the Pulmonary Artery.
3. Mixing up the location of the bicuspid and tricuspid valves; remember Tricuspid is on the Right and Bicuspid/Mitral is on the Left.
4. Using the terms Atherosclerosis and Arteriosclerosis interchangeably; Atherosclerosis is specifically about fatty plaque, while Arteriosclerosis is the general hardening of arteries.
5. Forgetting that the Left side of a heart diagram represents the patient's left, which is the right side of the paper for the viewer.
6. Incorrectly identifying the universal donor as O positive instead of O negative, or the universal recipient as AB negative instead of AB positive.

6. Practice Questions and Model Answers

Question 1

Explain the Cohesion-Tension Theory of water movement in plants.

Answer 1: The Cohesion-Tension Theory explains that water is pulled up through the xylem due to transpiration. As water evaporates from the leaves (transpiration), it creates a tension or pull. Because water molecules are cohesive (stick together) and adhesive (stick to xylem walls), they form a continuous, unbroken column from roots to leaves. This pull is strong enough to lift water to the tops of tall trees.

Question 2

Differentiate between Arteries and Veins based on structure and function.

Answer 2: Arteries have thick, elastic, and muscular walls to carry blood away from the heart under high pressure; they generally carry oxygenated blood (except the pulmonary artery) and do not have valves. Veins have thinner walls with less muscle, carry blood toward the heart under low pressure, generally carry deoxygenated blood (except pulmonary veins), and contain valves to prevent backflow.

Question 3

What is the ABO blood group system, and how is it determined?

Answer 3: The ABO blood group system classifies blood into four types: A, B, AB, and O. It is determined by the presence or absence of specific antigens (A and B) on the surface of red blood cells. Type A has antigen A, Type B has antigen B, Type AB has both, and Type O has neither. Corresponding antibodies (anti-A and anti-B) are found in the plasma.

Question 4

Describe the role of the Lymphatic System in human transport.

Answer 4: The lymphatic system acts as a secondary transport system that returns excess interstitial fluid (lymph) back to the circulatory system. It also transports absorbed fats from the small intestine to the blood via lacteals and plays a vital role in immunity by housing white blood cells in lymph nodes to filter out pathogens.

Question 5

What are the causes and symptoms of a Myocardial Infarction?

Answer 5: A myocardial infarction (heart attack) is caused by a blockage in the coronary arteries, usually due to a blood clot or atherosclerosis, which stops blood flow to the heart muscle. Symptoms include severe chest pain (angina), shortness of breath, sweating, and pain radiating to the left arm or jaw. It results in the death of the affected heart tissue.